

Rowan Ali Ibrahim Ali

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Profile:

Data Science student at Alexandria University seeking an entry-level Data Science or AI role where I can apply my strong foundation in machine learning, data analytics, NLP, forecasting, and distributed systems to solve real-world problems. I have developed end-to-end projects involving large-scale data processing, predictive modeling, medical text analytics, forecasting, and business intelligence applications, with a strong focus on transforming raw and complex data into meaningful insights and actionable solutions.

Proficient in Python, R, SQL, statistical analysis, and scalable data pipelines, with hands-on experience in data preprocessing, exploratory data analysis, feature engineering, machine learning, and data-driven problem solving. I also have additional exposure to deep learning, computer vision, database systems, software design, computer networks, and distributed computing, allowing me to approach data challenges from both analytical and technical perspectives.

Passionate about Data Science and Artificial Intelligence and continuously motivated to expand my technical knowledge, strengthen my problem-solving abilities, and apply data-driven approaches to meaningful challenges. I aim to contribute to innovative projects by combining analytical thinking, technical skills, and a strong willingness to learn, while delivering reliable, scalable, and practical data solutions that create measurable value for businesses and end users.

Education

Education

2023 – 2027

Bachelor of Science (B.Sc.) in Data Science

Faculty of Computing and Data Science, **Alexandria University**

Major: Data Science

Experience

1)MAIM Training — Artificial Intelligence & Machine Learning Trainee

Jul 2025 – Sep 2025 | Competitive, Project-Based Training Program

- Completed a competitive, project-based training program in **Artificial Intelligence and Machine Learning**, covering Machine Learning, Deep Learning, Neural Networks, NLP, Computer Vision, and MLOps through practical assignments and performance-based assessments.
 - Developed end-to-end machine learning solutions by performing **data cleaning, preprocessing, exploratory data analysis, feature engineering, model development, evaluation, and deployment**, using Python-based data science and machine learning tools.
 - Applied **NumPy, Pandas, Matplotlib, Seaborn, and Scikit-learn** to transform raw datasets into structured, analysis-ready data and generate meaningful visual and statistical insights.
 - Built and evaluated predictive models using **Linear and Polynomial Regression, Logistic Regression, Decision Trees, Naive Bayes, K-Nearest Neighbors, SVM, Ensemble Learning, and Clustering**, selecting appropriate approaches based on the problem and evaluation results.
 - Completed practical projects including **Titanic analysis, California Housing prediction, COVID-19 data analysis and visualization, and Breast Cancer Diagnosis Prediction**, strengthening analytical thinking, problem-solving, and attention to detail through real-world datasets.
 - Applied deep learning concepts including **Neural Networks, regularization, overfitting control, network optimization, TensorFlow, and model fine-tuning**, developing a stronger understanding of model generalization and performance improvement.
 - Worked on **NLP and Computer Vision** tasks involving text representation, RNNs, LSTMs, attention mechanisms, Transformers, CNN-based image classification, transfer learning, and object detection.
 - Applied deployment concepts using **Streamlit and Flask**, converting trained models into accessible applications and strengthening the transition from experimentation and model development to practical implementation.
 - Completed technical quizzes, assignments, and competitive tasks under performance-based evaluation, demonstrating **adaptability, fast learning, active listening, communication, and a competitive mindset** while continuously applying feedback to improve technical work.
 - **Recognized as a top participant in the program based on performance and assessment results**, demonstrating consistent technical performance, commitment, and the ability to rapidly learn and apply new concepts.
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2)Huawei ICT Academy — HCIA-Big Data Trainee

[August 2026] | Huawei ETA Academy & NTI

- Completed **HCIA-Big Data training**, developing a practical foundation in Big Data technologies, distributed data processing, and large-scale data management.
 - Studied core Big Data architecture and processing concepts, including **HDFS, MapReduce, YARN, Hive, HBase, Spark, Flink, Kafka, Flume, ZooKeeper, Elasticsearch, and Redis**, with emphasis on understanding how distributed systems process and manage large-scale data.
 - Applied distributed computing concepts to understand how data is **stored, processed, transferred, and managed across scalable environments**, strengthening analytical thinking and problem-solving skills in data-intensive scenarios.
 - Worked through technical exercises and assessments involving Big Data architecture and ecosystem components, demonstrating **attention to detail, adaptability, active learning, and the ability to understand complex technical systems**.
 - Successfully completed the training and its required assessments, strengthening my foundation in **Big Data, distributed systems, data engineering concepts, and scalable data processing**.
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3)FlyRank AI — Machine Learning Intern

June 2026 – Present | Remote

- Currently participate in a **project-based Machine Learning internship** focused on applying machine learning concepts to practical problems using real data and modern AI tools.
- Work through a structured, self-paced ML track that emphasizes **problem framing, data preparation, model development, evaluation, and building portfolio-ready machine learning solutions**.
- Apply machine learning concepts to practical assignments and projects, focusing on developing reliable models and interpreting their results rather than treating model training as a standalone task.
- Work with modern AI-assisted development workflows and technical tools to accelerate experimentation, problem solving, and implementation while maintaining a structured approach to evaluating outputs.
- Develop a portfolio-oriented **capstone project**, turning the internship work into tangible evidence of technical ability and strengthening practical experience in machine learning engineering.
- Incorporate feedback from reviews and practical work to refine solutions, demonstrating **adaptability, attention to detail, fast learning, and a continuous improvement mindset**.

- Work independently within a self-paced environment while managing technical tasks, deadlines, and learning objectives, strengthening **time management, problem-solving, self-direction, and communication**.
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4) Digital Egypt Pioneers Initiative (DEPI) — Machine Learning Intern

[July 2026] – Present | Egypt

- Currently completing a structured, hands-on **Machine Learning internship** covering mathematical foundations, Python programming, data preprocessing and visualization, machine learning, deep learning, NLP, computer vision, Azure, and MLOps.
 - Apply **Python, NumPy, Pandas, Matplotlib, and Seaborn** to perform data manipulation, preprocessing, exploratory analysis, and visualization through practical assignments and technical exercises.
 - Develop and evaluate machine learning models using **regression, classification, decision trees, Naive Bayes, KNN, SVM, ensemble learning, clustering, feature selection, and dimensionality reduction**, strengthening analytical thinking and systematic problem-solving.
 - Work with **neural networks, TensorFlow, regularization, overfitting control, optimization, and model fine-tuning**, applying theoretical concepts through practical machine learning tasks.
 - Apply NLP concepts including **text representation, sequential modeling, RNNs, LSTMs, attention mechanisms, and Transformers**, while developing the ability to process and model unstructured textual data.
 - Practice Computer Vision techniques including **image processing, CNNs, transfer learning, and object detection**, applying careful data preparation and model evaluation to visual learning tasks.
 - Explore **Microsoft Azure and MLflow** as part of the machine learning workflow, developing an understanding of cloud-based AI services and MLOps practices for more scalable and maintainable solutions.
 - Complete ongoing quizzes, assignments, and practical tasks while actively applying feedback, adapting to new technical concepts, and demonstrating **problem-solving, attention to detail, fast learning, active listening, and adaptability**.
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Technical Skills

Data Science & Analytics:

Data Analysis, Exploratory Data Analysis (EDA), Data Preprocessing, Data

Cleaning, Feature Engineering, Statistical Analysis, Predictive Modeling, Data Mining, Data Visualization, Business Intelligence, Forecasting, Web Scraping

Machine Learning & Artificial Intelligence:

Supervised Learning, Unsupervised Learning, Regression, Classification, Clustering, Ensemble Learning, Feature Selection, Dimensionality Reduction, Model Evaluation, Neural Networks, Deep Learning, Natural Language Processing (NLP), Computer Vision, Generative AI Fundamentals

Big Data & Data Engineering:

Big Data Processing, Distributed Computing, Scalable Data Processing, Data Pipelines, ETL Concepts, Distributed Storage, Batch Processing, Stream Processing, Data Processing Frameworks, Big Data Ecosystems, MLOps Fundamentals

Programming & Data Technologies:

Python, R, SQL, C++, Java, JavaScript, HTML, CSS; Object-Oriented Programming, Data Structures & Algorithms, Functional Programming Concepts, API Integration, JSON Data Processing

Data Platforms, Databases & Cloud:

Relational Databases, SQL Databases, MongoDB, Database Design, Data Modeling, Microsoft Azure, AWS Fundamentals, Cloud Computing, Distributed Databases

Machine Learning Engineering & Deployment:

Model Development, Model Evaluation, Model Fine-Tuning, ML Pipelines, Model Deployment, REST APIs, Application Integration, Streamlit, Flask, MLflow

Data Visualization & Business Intelligence:

Power BI, Matplotlib, Seaborn, Dashboard Development, Data Storytelling, KPI Analysis, Business Reporting

Software & Development Tools:

Git, GitHub, Jupyter Notebook, VS Code, API Development, Software Design, Version Control

Systems & Computer Technologies:

Distributed Systems, Computer Networks, Operating Systems Fundamentals, Client–Server Architecture, Software Engineering Fundamentals

Extracurricular & Volunteer Experience

HackerRank Family — Scientific Volunteer

Faculty of Computing and Data Science, Alexandria University

- Volunteered as a member of the **HackerRank Family**, contributing to scientific and academic activities within the Faculty of Computing and Data Science.
 - Prepared and shared **concise scientific summaries and educational content** to help students understand and review technical topics more effectively.
 - Created **explanatory videos** that simplified complex scientific and technical concepts into clear, structured, and accessible explanations.
 - Supported fellow students who faced difficulties with academic or technical topics by explaining concepts, discussing solutions, and providing peer-to-peer assistance.
 - Contributed to a collaborative learning environment, strengthening **communication, teamwork, active listening, problem-solving, and knowledge-sharing skills**.
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Safwa Family — Scientific Volunteer

Faculty of Computing and Data Science, Alexandria University

- Volunteered as a member of the **Safwa Family**, supporting scientific and academic activities and contributing to a collaborative student learning environment.
 - Developed and organized **scientific summaries and learning materials** to make technical content easier to understand, revise, and apply.
 - Assisted students with challenging scientific topics by providing explanations, clarifying concepts, and helping them approach problems through structured reasoning.
 - Contributed to **educational explanation videos and peer-learning activities**, adapting explanations to different levels of understanding.
 - Strengthened **communication, teamwork, adaptability, initiative, patience, and problem-solving skills** through continuous peer support and scientific volunteering.
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Languages

Arabic: Native

English: Upper-Intermediate (B2+/C1 level)

Projects

1) Distributed Weather Forecasting & Analytics Platform

Python | Distributed Systems | Data Engineering | Forecasting

GitHub: <https://github.com/Rowan-ali/demand-forecasting-mlops>

- Designed and developed a **distributed weather forecasting and analytics platform** for processing, analyzing, and forecasting large-scale weather data.
 - Engineered a scalable data workflow incorporating **data partitioning and replication** to improve scalability, reliability, and fault tolerance across distributed processing tasks.
 - Built an end-to-end data pipeline covering **data preprocessing, transformation, feature engineering, analytical processing, and forecasting**.
 - Developed forecasting models to predict future weather conditions and evaluated model performance using appropriate forecasting metrics.
 - Applied distributed computing principles to efficiently process and analyze **large volumes of weather data**, reducing reliance on centralized processing.
 - Integrated distributed data processing with predictive analytics into a unified forecasting workflow, demonstrating practical application of **data engineering, distributed systems, and machine learning concepts**.
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2) Medical NLP & Semantic Disease Analysis

Python | NLP | TF-IDF | LSA | SVD | SciSpacy

- Developed an end-to-end **medical NLP and semantic analysis pipeline** for extracting disease-related knowledge and hidden semantic patterns from **18,000+ biomedical and clinical research documents**.
 - Engineered the text-processing pipeline through **text cleaning, tokenization, POS tagging, custom stop-word removal, lemmatization, and feature engineering** to transform unstructured medical text into structured representations.
 - Applied **TF-IDF vectorization** and refined the vocabulary to **5,000 high-value biomedical features**, improving the representation of domain-specific medical terminology.
 - Applied **Latent Semantic Analysis (LSA)** using Singular Value Decomposition (SVD) to reduce dimensionality and uncover hidden topics and semantic relationships within biomedical literature.
 - Implemented **Biomedical Named Entity Recognition (NER) with SciSpacy** to identify clinically relevant entities including diseases, genes, chemicals, cells, and anatomical structures.
 - Developed interactive visualizations and semantic exploration tools to support **medical topic discovery, disease-related knowledge extraction, and downstream analysis**.
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3) Crime Trend Prediction System

Python | Machine Learning | Data Mining | SQL

GitHub: <https://github.com/Rowan-ali/Crime-Rate>

- Developed a **crime trend prediction system** to analyze historical crime patterns and forecast future trends using **500,000+ crime records**.
 - Built a robust data-processing workflow covering **data cleaning, preprocessing, transformation, feature engineering, and exploratory data analysis** to prepare large-scale crime data for predictive modeling.
 - Conducted exploratory analysis to identify **temporal, regional, and behavioral crime patterns**, transforming raw records into meaningful analytical insights.
 - Developed and compared machine learning models for **crime trend prediction**, selecting appropriate approaches based on validation results and predictive performance.
 - Integrated **SQL-based data management** to support structured storage, retrieval, and analysis of crime-related information.
 - Evaluated predictive approaches using appropriate **performance metrics and validation techniques**, supporting data-driven interpretation of future crime trends.
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4)Retail Analytics & Market Basket Analysis

R | Statistical Analysis | Association Rule Mining | Business Intelligence

- Developed an end-to-end **retail analytics solution** using R to analyze supermarket transaction data and uncover customer purchasing behavior.
 - Performed **data cleaning, preprocessing, exploratory data analysis, and statistical analysis** to identify sales patterns and customer trends.
 - Applied **Association Rule Mining and Market Basket Analysis** to discover relationships between products and identify frequently occurring purchase combinations.
 - Developed analytical models and data-driven insights to support **sales optimization, product placement, and business decision-making**.
 - Applied statistical techniques and data visualization to identify **customer behavior patterns and operational opportunities**.
 - Translated analytical findings into **actionable business recommendations and structured reports**, connecting technical analysis with practical business value.
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